

Individual Work Breakdown Report

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Project Context

This report documents my individual contribution to the group project “Dublin Housing Affordability Analysis (2015–2025).” The project objective was to design and implement an end-to-end analytics workflow integrating structured and semi-structured data, storing outputs in both non-relational and relational databases, and communicating findings through static and interactive visualizations.

Primary Ownership Areas

- Source identification, validation, and acquisition planning.
- CSV ingestion and raw persistence architecture for the property dataset.
- MongoDB raw-zone schema design and insertion logic.
- Cross-stage quality review and integration validation.

Detailed Contributions Across the Analytics Lifecycle

1. Data Selection and Collection

I led source validation for the project by identifying suitable non-Kaggle datasets that aligned with the assignment constraints. I checked source trustworthiness, data format relevance, and reuse feasibility. I supported decisions to use the Property Price Register as the transaction base and CSO APIs as semi-structured socioeconomic context.

2. Extraction and Raw Data Engineering

I implemented and tested key extraction components for CSV ingestion, including path checks and load validation. I also worked on raw persistence controls in MongoDB so that source records and payload-level snapshots are retained before transformation. This was important for reproducibility and auditability, particularly for API data that may change schema over time.

3. Database Design and Storage Responsibilities

I contributed to designing raw-zone collection structure in MongoDB:

- `property_prices_raw`
- `cso_income_raw` and `cso_income_parsed_raw`
- `cso_population_raw` and `cso_population_parsed_raw`

I verified insertion behavior and collection refresh strategy across multiple runs to ensure deterministic outputs.

4. Transformation and Integration Participation

Although not my primary ownership, I participated in reviewing transformation rules and integration joins. I tested whether extracted records were correctly handed into cleaning and aggregation stages and validated that transformations did not break raw-to-curated lineage.

5. Validation and Quality Checks

I executed repeatability checks after pipeline runs, including:

- raw and processed file existence checks,
- collection/table population verification,
- sanity checks on year ranges and output table structure.

I also helped verify that changes in parsing logic did not silently degrade downstream outputs.

6. Reporting and Communication

I co-authored sections related to architecture rationale, data acquisition, and reproducibility. I contributed to preparation of the final narrative used in presentation rehearsal, especially sections explaining raw-zone design and why source lineage matters for data analytics credibility.

Cross-Stage Evidence of Participation

- **Selection:** validated sources and project-data fit.
- **Collection:** implemented and verified extraction steps.
- **Storage:** owned raw persistence design in MongoDB.
- **Processing:** reviewed integration and transformation handoff quality.
- **Analysis:** participated in output sanity checks and interpretation discussions.
- **Visualization/Communication:** supported report and presentation framing.

Challenges and How I Addressed Them

Challenge 1: Maintaining reproducibility when source formats differ. **Action:** Promoted strict raw snapshot storage before transformation to preserve lineage and support replay.

Challenge 2: Avoiding brittle ingestion behavior for large CSV and API inputs. **Action:** Implemented validation checks and reviewed logging outputs after each run.

Challenge 3: Coordinating responsibilities while preserving full-lifecycle participation. **Action:** Maintained primary ownership in extraction/raw storage while actively participating in review and testing stages throughout the project.

Reflection

My main contribution was making the pipeline dependable at its entry point. I focused on source integrity, extraction robustness, and raw data traceability, which are foundational for trustworthy analytics. This work strengthened reproducibility and reduced risk when later transformation and visualization layers evolved.

Declaration

This report is an original account of my own contribution to the team project.